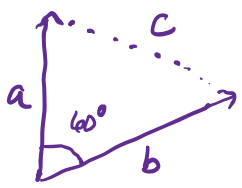
Problem 11. TIER 2 — PYTHAGORAS IS THE SPECIAL CASE

Two ships leave port at the same time on courses 60° apart, one at 15 km/h and the other at 20 km/h. How fast is the distance between them growing after 2 hours?

(1) 

(2) $\frac{da}{dt} = 15$, $\frac{db}{dt} = 20$, want $\frac{dc}{dt}$ when $t = 2$

(3) $c^2 = a^2 + b^2 - 2ab\cos 60°$ ($\cos 60° = \frac{1}{2}$)

$c^2 = a^2 + b^2 - ab$

(4) $\frac{d}{dt}c^2 = \frac{d}{dt}(a^2 + b^2 - ab)$

$2c\frac{dc}{dt} = 2a\frac{da}{dt} + 2b\frac{db}{dt} - \left(a\frac{db}{dt} + b\frac{da}{dt}\right)$

(5) at $t = 2$ hrs, $a = 2(15) = 30$, $b = 2(20) = 40$

$c^2 = 30^2 + 40^2 - 30\cdot 40 = 50^2 - 30\cdot 40 = 2500 - 1200 = 1300$

$c = \sqrt{1300} = \sqrt{13}\sqrt{100} = 10\sqrt{13}$

$2\cdot 10\sqrt{13}\frac{dc}{dt} = 2\cdot 30\cdot 15 + 2\cdot 40\cdot 20 - (30\cdot 20 + 40\cdot 15)$

$20\sqrt{13}\frac{dc}{dt} = 900 + 1600 - 600 - 600 = 1300$

$\frac{dc}{dt} = \frac{1300}{20\sqrt{13}} = \frac{65}{\sqrt{13}}\frac{\sqrt{13}}{\sqrt{13}} = \frac{65\sqrt{13}}{13} = \frac{5\cdot 13\sqrt{13}}{13}$

$= 5\sqrt{13}\ \frac{\text{km}}{\text{hr}} \approx 18.0\ \text{km/hr}$